DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

UNITED NATIONS DEVELOPMENT PROGRAMME

Empowered lives.
Resilient nations.

PROJECT DOCUMENT

[State of Palestine – Gaza Strip]

Project Title: **Solar Energy System for Khan Younis Waste Water Treatment Plant**

Project Number: Award: PAL 10-00121913, Output: PAL 10- 00117719

Implementing Partner: UNDP/PAPP

Start Date: 1 December 2019 **End Date:** 31 December 2022 **PAC Meeting date:** 23 January 2020

Brief Description

Khan Younis Waste Water Treatment Plant (KY WWTP) is a strategic environmental project that aims to improve access to adequate, inclusive and equitable sanitation and hygiene for 389,000 residents of Khan Younis Governorate with special attention to the needs of women. It will broadly contribute to the protection of the public health and environment in the Gaza Strip.

Whilst the operation of KY WWTP started in November 2019, the provision of the electrical power represents a major and immediate challenge within the protracted and worsening shortage of electricity in Gaza. To provide a constant supply of electricity for the plant, the Palestinian Energy and Natural Resources Authority (PENRA) has endorsed the proposed intervention for implementing an on-grid solar energy system with a capacity of 07 MWp for KY WWTP that can be designed as an extendable system and constructed into integrated phases.

The overall project of the solar energy system with a capacity of 07 MWp aims to achieve optimal operation of KY WWTP and reduce the overall running cost by around 50%, helping to attain a sustainable operation of the plant. It comes in line with the UNDP's strategy for safeguarding the environment and protecting natural resources and responds to the 10th national priority of the National Policy Agenda (NPA) resilient communities: ensuring a sustainable environment and adapting to climate change; policy intervention: increase energy efficiency and reliance on renewable energy.

Within fund from the Government of Norway amounting to US$ 2,179,836.51, the detailed design for the 07 MWp solar energy system for KY WWTP will be provided and the first phase with a capacity of 1.3 MWp will be constructed, which will be followed by constructing the complementary phases upon the availability of the required funds.

The awareness of Khan Younis residents especially women regarding the importance of clean water and hygiene will be improved, and the engagement of communities in plans for effective utilization of the treated water will be increased.

Contributing Outcome (UNDAF/CPD, RPD or GPD):

UNDAF Outcome 3.3: Palestine's infrastructure, and natural and cultural resources are more sustainably used and managed.

Indicative Output(s) with gender marker[2]:

CPD Output 3.6: Installation of solar panels and mainstreaming solar energy supported.

Gender Marker: GEN2 (Gender equality as a significant objective).

Total resources required:	US$ 10,000,000	
Total resources allocated:	US$ 2,179,836.51	
	UNDP TRAC:	
	Donor: Norway	US$ 2,179,836.51
	Government:	
	In-Kind:	
Unfunded:	US$ 7,820,163.49	

Agreed by (signatures)[1]:

UNDP
Yvonne Helle Special Representative of the Administrator UNDP/PAPP
Date:

[1] Note: Adjust signatures as needed

[2] The Gender Marker measures how much a project invests in gender equality and women's empowerment. Select one for each output: GEN3 (Gender equality as a principle objective); GEN2 (Gender equality as a significant objective); GEN1 (Limited contribution to gender equality); GEN0 (No contribution to gender quality)

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

I. DEVELOPMENT CHALLENGE

The availability of fresh water in Palestine is amongst the lowest in the world. The Gaza Strip is located in a semi-arid region and lacks natural water resources. Groundwater from the coastal aquifer is the unique water source available for drinking, domestic and agriculture purposes in the Gaza Strip. Because of increased demand, groundwater is being extracted from this aquifer at almost three and a half times the sustainable abstraction rate, thereby causing seawater intrusion. Moreover, sewage pollution through cesspits contributes to making the water aquifer largely non-potable. The Palestinian Water Authority (PWA) assessment report on water quality, issued in January 2017 confirms that 96.5% of the aquifer water is yielding water that fails all drinking water quality standards. The UN report "Gaza in 2020: a liveable place?" indicated that Gaza aquifer would be completely unusable by 2016 and irreversibly damaged by 2020, should there be no interventions to save the aquifer.

The waste water sector in the Gaza Strip is suffering from the shortage of functional disposal and treatment facilities. Despite 85 percent of the population having sewer-connections, there are many areas and cities, especially in Khan Younis Governorate, that have no sewage system at all. The population of these areas are still relying on cesspits that cause serious contamination to the water aquifer. The longstanding shortage of such functional facilities contributes to the discharge of 100-108 million litres of poorly treated sewage into the sea every day. This situation poses serious health and environmental hazards, particularly during the summer, when swimming in the sea is one of the few recreational activities available to the population of Gaza. According to WHO, water-related diseases account for over one quarter of illnesses and are the primary cause of child morbidity in the Gaza Strip. The Environmental Quality Authority in Gaza declared the beaches of Gaza City, the main destination for internal tourism, and in Rafah governorates, as highly polluted. These beaches constitute 75 per cent of all Gaza beaches and swimming in these areas is strongly discouraged.

The electricity shortage is another major cause undermining the provision of water, sanitation and hygiene services. Gaza has been suffering from a chronic electricity deficit since 2006, which was aggravated during the 2014 hostilities when the Gaza Power Plant (GPP) was destroyed. While electricity needs are around 450 MW, only around 85 MW is produced by the GPP, electricity supplied from Egypt is non-existent, and supply from Israel is around 120 MW, leaving a gap of around 245MW. Moreover, fuel costs to operate the generators are amounting to approximately USD 10 million on a monthly basis, with a yearly investment of around USD 150 million on average. Continued reliance on humanitarian fuel is not sustainable. The dire economic situation in Gaza has also led to very a low collection rate of electricity fees, placing the burden of covering extra fuel costs on Gaza Electricity Distribution Corporation (GEDCO), which is also facing severe capacity gaps. Interruptions to electricity is not a matter of inconvenience, but severely affects delivery of essential services.

The Khan Younis Governorate extends over a wide geographic area of around 108 square kilometres in the southern part of the Gaza Strip, and it is inhabited by around 389,000 residents (projected population in 2019, based on the 366,000-total population in 2017, PCBS). The Governorate suffers from an absence of a comprehensive sewerage collection system and functional waste water treatment plant. To face the exacerbated sewage disposal problems and dire hygienic conditions in crowded areas, Khan Younis and Bani Suhaila Municipalities managed to serve part of their residents by a public sewage collection system. At present, only 40% of the population of Khan Younis Governorate are served by this sewerage system. The collected sewage was diverted to six ad-hoc lagoons established temporarily in 2008 in the western side of Khan Younis. Up to September 2019, more than 15,000 cubic metres of partially treated wastewater was discharged through these lagoons to the Mediterranean Sea. The sewage generated from unserved areas is still being evacuated from cesspits by vacuum trucks and disposed of in the environment without any treatment. This is posing serious risks to public health as well as contaminating the ground water aquifer, sea water and marine life.

Khan Younis Waste Water Treatment Plant (KY WWTP) is prioritised by the PWA as a strategic environmental project. It aims to protect the public health, water resources and environment of Khan Younis residents, through the development of public and social infrastructure in the Gaza Strip. Women are among the main beneficiaries from the services of KY WWTP. Of the 259,000 residents of Khan Younis and Khan Younis Camp who will benefit directly from the project, 49.3% (PCBS, Feb. 2018) are women who will enjoy decreased daily life stressors due to improved access to adequate waste water collection and treatment and the subsequent improvement in the public health conditions including decreased pollution of ground water, sea water and marine life and decreased water-borne diseases. It will contribute to achieving the PWA's objective for water security, and balanced water management by providing a new non-conventional water source of treated waste water that will recharge the aquifer that can be used for agricultural purposes. UNDP/PAPP developed the detailed design for the construction of KY WWTP. The implementation plan was developed by designing an extendable waste water treatment plant to be constructed in two phases

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

according to available resources. The design figures of KY WWTP, Phase I, are based on flow of 26,600 cubic metres per day, while the design figures of Phase II are based on flow of 44,900 cubic metres per day.

With funding from the Government of Japan, the Kuwait Fund for Arab Economic Development through the Islamic Development Bank, and co-financing by UNDP, USD 58 million has been allocated for the construction of KY WWTP, Phase I. This entails the first phase of the treatment plant, effluent and emergency pressure pipeline, infiltration basins to recharge the treated effluent into the aquifer and its electrical power supply line. This phase is currently under implementation by UNDP/PAPP. The operation stage of the plant started after completing the commissioning stage in early November 2019.

While Phase I includes costs for the contractor to operate the plant for one year after commissioning, as well as technical knowledge transfer to the operator, it does not include the costs of the electricity, consumables and the operator staff required for the operation and maintenance. Due to the huge electricity deficit, the KY WWTP will rely heavily on back-up generators run on fuel to operate at a limited capacity as the cost for 1kWh of electricity supplied by diesel generation is equivalent to minimum 2,3 kWh supplied by the main grid, the operator of KY WWTP, clarified that they could provide fuel to operate the diesel generators - as standby for emergency cases only.

Whilst the operation of KY WWTP started in November 2019 as anticipated, the provision of the electrical power represents a major and immediate challenge within the protracted and worsening shortage of electricity in Gaza. Reliable and uninterrupted electrical power is needed to maintain the biological treatment processes for the waste water treatment plant and is a prerequisite for a fully functional and effective operation of KY WWTP. It is therefore very important, in view of reducing the energy costs, that the percentage of electricity received from the main grid increases to at least 20 hours/day, and that an alternative power supply source from Photovoltaic panels, is implemented as soon as possible.

UNDP conducted consultation meetings in November and December 2018 with the concerned parties including the PWA, the CMWU, Khan Younis Municipality, the PENRA and GEDCO to discuss and agree upon a practical approach to secure continuous electricity needed to operate KY WWTP.

PENRA highlighted that increases in power are not expected in Gaza before 2021. The CMWU, the operator of KY WWTP, clarified that they could provide fuel to operate the diesel generators - as standby for emergency cases only. PENRA and all parties agreed that the construction of a solar energy system would be the best option to support GEDCO in meeting the urgent electrical demands for KY WWTP.

II. STRATEGY

While Gaza has been suffering from a chronic electricity deficit since 2006, the situation significantly worsened during the 2014 hostilities when the Gaza Power Plant (GPP) was destroyed. Moreover, there is increasing donor fatigue regarding the continuation of funding for fuel to operate the generators, which was initially intended to be a temporary measure. The severe economic situation in Gaza has also led to very a low collection rate of electricity fees which GEDCO relies on. These conditions have placed constraints to the operation of KY WWTP that were not initially anticipated during the design and construction of the plant.

Due to the dire economic and political situation, and CMWU having limited resources, they have indicated that they are not able to cover the full costs associated with the running of KY WWTP. As electricity comprises a major part of the operational costs, a more cost-efficient system is required to secure immediate and sustainable operation of the plant.

Solar renewable energy has the highest potential to add new resource to the electricity grid due to Gaza's sunny weather, which lasts for more than 300 days annually. The Palestinian government has recognised the importance of opportunities in renewable energy, and in 2011, a national strategy was launched for mainstreaming renewable energy as a new source for providing electricity services in Palestine. PENRA is aiming to achieve 25% of energy demands in the State of Palestine from renewable resources by 2020.

In November 2015, UNDP carried out a preliminary study to assess the solar energy needed to operate the plant and proposed scenarios. The energy consumption of KY WWTP, Phase I, will be around 12.6 GWh per year and thus a 7.3 MWp PV plant would produce adequate power, provided that there is sufficient energy storage capacity to ensure match between consumption and electricity delivery. This will allow the surplus of the power generated during the day from the solar energy plant to be stored in the main grid and to be returned to KY WWTP at night. Thus, electricity can be provided on a continuous basis (24/7) for KY WWTP. The implementation plan for the 07 MWp plant is based on designing an extendable solar energy system to be constructed into integrated phases. The detailed design for the 07 MWp solar energy system will be developed while constructing the first phase of this solar energy system with a capacity of 1.3 MWp.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

The electrical line to connect the plant with the GEDCO main grid has already been constructed through the UNDP intervention with KY WWTP. The electrical power from GEDCO main grid has been successfully introduced to KY WWTP site for testing purposes in May 2019. In its letters to UNDP, PENRA indicated that the electrical power will be provided on a commercial basis to KY WWTP in conjunction with the implementation of the solar energy project to KY WWTP.

The proposed intervention for implementing the solar energy project for KY WWTP is expected to provide a constant supply of electricity and reduce the overall running cost by around 50%, helping to attain a sustainable operation of the plant.

The functional and sustainable operation of KY WWTP will stop disposing off the raw sewage into cesspits and the Mediterranean Sea that will contribute to minimizing pollution to water aquifer, sea water and marine life and the related waterborne diseases. This will eventually contribute to improving the environment, public health and wellbeing of the 389,000 residents of Khan Younis Governorate.

Within the full operation of KY WWTP, Phase I, at a capacity of 26,600 cubic metres per day, more than 9 million cubic meters of treated waste water will annually be recharged into the poor local aquifer, thus, providing a new non-conventional water source to be used for agricultural purposes in the extended agricultural lands at the eastern parts of Khan Younis. This will contribute to lessening the pressure on the Gaza's deteriorated water aquifer, enhancing the agricultural productivity in Khan Younis area and improving the livelihood of farmers.

Job opportunities will be generated for unemployed local workers that will contribute to promoting their socio-economic conditions through a dignified livelihood.

III. RESULTS AND PARTNERSHIPS

Expected Results

- Through the operation of KY WWTP utilizing environmentally friendly renewable solar energy, this project responds to the chronic humanitarian needs in Gaza. It contributes to the United Nations Sustainable Development Goals (SDG's) related to ensuring conservation of environment, including access to water, sanitation, energy and gender equality (SDG 3, 5, 6, 7, 11, 13, and 14). It is also in line with UNDP's strategy for safeguarding the environment and protecting natural resources. This project is also aligned with the national policy agenda of 2018-2022 where national priority has a clear output on environment "Sustainable Environment and Adapting to Climate Change".

  The project aims to:

  - Improve the public health and wellbeing, and the resilience of Palestinians in Khan Younis especially vulnerable groups such as women and children and including refugees.
  - Ease the pressure on the national electricity grid.
  - Contribute to preserving the environment by lowering Co2 emissions and safe guarding the aquifer.

Level	Expected Results
Impact	Improved access to adequate, inclusive and equitable sanitation and hygiene for residents of Khan Younis Governorate with special attention to the needs of women though the optimal operation of the Khan Younis Waste Water Treatment Plant (KYWWTP) utilizing renewable sustainable photovoltaic solar energy, contributing to the protection of the public health and environment in the Gaza Strip.
Outcome	Palestine's infrastructure, and natural and cultural resources are more sustainably used and managed.
Output 1	07 MWp Renewable solar energy installed in KYWWTP to sustainably treat the wastewater in Khan Younis governorate.
Output 2	Improved awareness of Khan Younis residents especially women regarding the importance of clean water and hygiene, and increased engagement of communities in plans for effective utilization of treated water.

Resources Required to Achieve the Expected Results

- To achieve the expected results, UNDP will manage the project and directly implement and supervise the different activities in close coordination with partners. For that, UNDP will assign a Programme

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

Analyst/ Programme Manager to monitor and follow-up on all managerial issues, including administrative and financial aspects related to the project, and ensure the quality of the project throughout the implementation process. UNDP will also assign site and electrical engineers who will be responsible for day-to-day supervision and monitoring of the technical aspects of the project.

The overall budget required for the implementation of the overall intervention of 07 MWp solar energy system for KY WWTP is estimated at around US$ 10 million. To date, UNDP/PAPP has secured US$ 2,179,836.51 from Norway for the implementation of a solar energy system with a capacity of around 1.3 MWp. This provides a total of around 21.8% of the overall required funding.

The solar energy system of 1.3 MWp capacity to be constructed within the fund from Norway is anticipated to act as the first phase of the overall solar system intervention. This would allow for future expansion to cover the full requirements (07 MWp) for the operation of the KY WWTP: an on-grid solution to ensure continuous operation.

The investment cost required to design and construct the 07 MWp solar energy system including the 1st phase of it with a capacity of around 1.3 MWp is given in the below table:

Item	Cost Estimate US$			
	1st phase capacity of 1.3 MWp	**2nd phase capacity of 3.6 MWp**	**3rd phase capacity of 2.1 MWp**	**Total Cost**
Comprehensive Design of extendable solar energy system with a capacity of 07 MWp for KY WWTP	120,000.00	0.00	0.00	120,000.00
Supply and install an on grid solar energy system with a capacity of 07 MWp	1,393,467.00	3,800,000.00	2,140,000.00	7,333,467.00
Construct infrastructure including fence, transformers & invertors rooms, interlock roads, etc.	221,710.00	450,000.00	175,000.00	846,710.00
UNDP direct implementation cost/ management and site supervision	210,868.00	283,791.71	270,408.41	765,068.12
Awareness and advocacy	52,338.31	50,000.00	0.00	102,338.31
Sub-total	**1,998,383.31**	**4,583,791.71**	**2,585,408.41**	**9,167,583.43**
GMS 8%	159,870.66	366,703.34	206,832.67	733,406.67
Coordination Levy	21,582.54	49,504.95	27,922.41	99,009.90
Total	**2,179,836.51**	**5,000,000.00**	**2,820,163.49**	**10,000,000.00**

Partnerships

- Partnership and coordination with PWA, PENRA, GEDCO, CMWU, Khan Younis Municipality and private sector will be maintained throughout the project. The respective roles are outlined below.
- PWA:
    - To coordinate and follow up with the Land Authority, Khan Younis Municipality and all concerned authorities to formally finalize the land acquisition issue and hand over the site for construction of the solar system project to UNDP/PAPP free of obstacles.
    - As the regulator of the water sector, to obtain the permit for connection of the solar system to the grid and signing the commercial agreement with GEDCO, to be responsible for covering the costs of energy consumption in accordance with the net metering system.
    - To sign the operation and maintenance agreement for the solar energy project with CMWU and Khan Younis Municipality.
- PENRA:

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

  - To give technical quality guidelines for the planned solar system.
  - To assist in implementing grid impact study for the solar system.
  - To assist in obtaining the appropriate net metering agreement between PWA and GEDCO.
  - To review the tender documents to ensure compliance with PENRA and GEDCO rules and regulations in connection with solar energy project.
  - To monitor the solar system after implementation and resolve any issues. (PENRA will assist in identifying any malfunction in the solar system that might arise, advise the probable solution and follow up on the implementation of the rectifications.)
- GEDCO:
  - To review the tender documents to ensure compliance with PENRA and GEDCO rules and regulations in connection with solar energy project.
  - To make commercial agreement with the operator of KY WWTP and the solar energy projects to store the surplus of the power generated from the solar energy system based on the net metering system.
  - To support UNDP in monitoring and supervising the onsite works of the solar system through the implementation stage.
- CMWU:
  - As service provider and final operator of the treatment plant, CMWU will be fully responsible for project management, operation and maintenance after construction and commissioning. CMWU's role in operating the KY WWTP will include the operation and maintenance of the new PV solar system that will be implemented and connected to KY WWTP facilities as well as the utility network.
- Khan Younis Municipality:
  - Work very closely with the PWA, CMWU, Palestinian Land Authority, all concerned authorities and land owners to secure the required land for the new PV solar system and the construction permits;
  - Play a key role with UNDP in responding to the residents' concerns;
  - Support in presenting the project objectives to the residents of Khan Younis Governorate;
  - Participate at the project board as senior beneficiary of the project;
  - Work with the CMWU collaboratively to deliver the required services to the residents of Khan Younis Governorate.

Risks and Assumptions

- Several external and internal risks have been identified in carrying out the activities of this project. These are registered in the Project Risk Log in Annex 2.
- The strong commitment of relevant authorities is critical in this politically volatile climate. The importance and priority for this solar energy system has been confirmed by PENRA and PWA. PENRA has endorsed the implementation of an on-grid solar energy project for KY WWTP with a capacity of 07 MWp that can be connected to the GEDCO main grid.
- Securing of the land required for the solar energy system presented a potential high-level risk, as the location is close to the border. Therefore, UNDP/PAPP mitigated early for this political and strategic risk and intensified efforts with the concerned authorities to assure acquiring the land required for the whole 07 MWp capacity project. The General Secretariat of the Cabinet of Gaza informed the Land Authority in May 2019 about their decision to acquire 80 dunums of land to be owned by Khan Younis Municipality and allocated for the solar energy project. The acquired land is located directly on the eastern side of KY WWTP site. The Chairman of the Land Authority has also requested Khan Younis Municipality in July 2019 to review the general administration of state property to complete the ownership and allocation procedures according to relevant regulations. In Sept. 2019, COGAT gave their support to solar PV developments at the proposed site. In November 2019, the Land Authority and Khan Younis Municipality negotiated the representatives of the land owners who accepted the compensation by exchanging their lands with other equivalent governmental ones and requested to establish an evaluation committee from governmental experts and the owners for that. However, and if this land is not secured for any reason, there is sufficient land within the KY WWTP premise to implement the 1.3 MWp first phase solar system instalment.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

Stakeholder Engagement

- The key partners outlined above will be closely involved in the design and implementation of the project through attending regular progress meetings, thus, ensuring that they have access to and are aware of the details to submit concerns about the technical, social and environmental impacts of the project.
- The tasks and functions of PENRA include developing the energy strategy, developing national plans and general policies related to the renewable energy sector in addition to developing and adopting specifications, standards and technical instructions related to this sector. Thus, and in accordance with the Exchange of Letters signed between UNDP and PENRA on 24 November 2019, PENRA will be responsible on the supervision of the implementation by participating in the official committee formulated by UNDP; this committee will be responsible for the progress report and final handover.

Gender and Women's Empowerment

- The results framework includes outputs and activities with clear gender focus to ensure that women equally benefit from the project as principal target group.
- For women, the shortage of fresh water means increased vulnerability. As the primary-care takers of household and family affairs, women are the ones to care for the children affected by water-borne diseases. Due to patriarchal, social norms and the effects of military conflict, they will also face additional obstacles in accessing sufficient health care for themselves and their children, especially in female-headed households. The lack of sufficient services – water as well as electricity – creates increased burdens on women in the household, both in terms of time-consuming mitigating routines as well as increased workload due to diseases and other consequences. This in turn keeps women far away from the labour market and decision-making positions. As an example, today's unemployment among Gazan women is at a staggering high with 62.8% in the first quarter of 2019 (PCBS, Labour force survey Q1/2019).

Knowledge

- UNDP/PAPP will work in close coordination with the national partners in promoting the project's components and its impact on the targeted population, and the Gaza strip at large.
- Throughout the project, UNDP/PAPP will ensure wide outreach on project's key components and the actual results/ impact achieved via extensive use of social and traditional mass media including publishing success stories, factsheet, press release, photographs, video and documentary.
- To exchange knowledge and proper codes of practice to the local operator, the contractor who will construct the solar energy system will operate it for a half year after commissioning within the engagement of the operator's staff.
- Through this project, UNDP will have an opportunity to promote women's economic empowerment in sectors that are traditionally male dominated. The project will target female fresh graduate engineers through a training seminar to strengthen their practical skills in the design and implementation of solar energy systems. This will enhance their participation in the labour market, especially as very few females are engaged in the electrical engineering sector.
- Throughout the project, a clean water and hygiene awareness raising activities for Khan Younis residents in addition to awareness for women on the best practices of hygiene and water-related issues will be conducted. The awareness plan will be implemented through UNDP implementing partner who has relevant expertise with WASH activities and have good outreach with different civil society organizations and communities in Khan Younis governorate as well as farmers who can benefit from the treated waste water for irrigating their crops in the agricultural lands located at the eastern side of Khan Younis Governorate. The implementing partners will be encouraged to engage the Mukhtarats of Gaza for interactional awareness for women. The activities will include surveys through questionnaires, interviews with stakeholders and community leaders and workshops.

Communications and Visibility

- Implementing the solar energy system for KY WWTP will have significant impact on the immediate and sustainable operation of such strategic and large-scale project and provide great visibility to the Government of Norway and UNDP.
- The communications and visibility activities will form part of the overall work plan of the project, with outcomes and achievements inherently built into the project design, reinforcing the values under which the project was implemented in the first place. All communications and visibility materials related to the project, as well as events and activities, will equally display the Government of Norway

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

and UNDP's logos, and in some cases the national counterpart's logo, in order to promote ownership and the strategic partnership between the different institutions.

- As mentioned above, success stories, factsheet, press release, photographs, video and documentary will be used to promote the project and crate visibility for knowledge sharing and lessons learned so others can benefit.
- All knowledge and communication materials will be produced in line with UNDP and donor's communication and visibility guidelines.

Sustainability and Scaling Up

- UNDP/PAPP will closely work with PWA, PENRA and Khan Younis Municipality who will ensure, from their complementarity mandates and professional experience, the realization of the project's outcomes and outputs.
- The General Secretariat of the Cabinet of Gaza issued a decision in May 2019 to acquire 80 dunums of land to be owned by Khan Younis Municipality and allocated for the solar energy project. The Chairman of the Land Authority has also requested Khan Younis Municipality in July 2019 to review the general administration of state property to complete the ownership and allocation procedures according to relevant regulations.
- PWA will make the operation and maintenance agreement for the solar energy project with CMWU and Khan Younis Municipality that is fundamental for the sustainability of this project. Through this agreement, the final operator who will allocate the required resources and take all charges for the continuous operation and maintenance of the project will be defined by PWA.
- To seek sustainability, the following is considered to enable CMWU effectively operate the solar PV system:
  1. The Scada system and necessary measurement tools required for proper operation and maintenance of the solar PV system will be supplied through the intervention.
  2. Complete cleaning system will be in place to minimize the operation cost and keep up the efficiency of the PV panels.
  3. Training on the proper operation and maintenance on the solar PV system will be provided for the operation team by the contractor.
  4. The contractor who will construct the solar energy system will operate it for a half year after commissioning the solar energy system to exchange knowledge and proper codes of practice to the operator.
- PENRA will monitor the solar system after implementation and resolve programming challenges. It will assist in identifying any malfunction in the solar system that might arise, advise the probable solution and follow up on the implementation of the rectifications, and will contribute to attaining sustainable operation of the project.
- The training to be provided by the contractor for the designated operator on the proper operation and maintenance of the solar energy system will contribute to the successful operation and sustainability of this project.
- UNDP/PAPP is vigorously working on identifying more funds to implement the complementary phases of the 07 MWp solar energy system to reduce 50% of the overall operational costs, and the operator of KY WWTP will not need to incur significant expenses for electricity consumption.

IV. PROJECT MANAGEMENT

Cost Efficiency and Effectiveness

- **Efficiency:** UNDP/PAPP's comparative advantage rests on its development approach of empowering the Palestinian people and their institutions to achieve a resilient Palestinian nation-based on national ownership and capacity development towards sustainability. With a long-standing presence in the occupied Palestinian territory, UNDP/PAPP has acquired a deep understanding of the local context. Being a trusted partner and convener, UNDP/PAPP enjoys a close partnership with Palestinian institutions, civil society, communities, academia, the United Nations Country Team (UNCT) and international development partners. Since its establishment in 1978, donor support to UNDP/PAPP has increased steadily reaching USD 1.7 billion. UNDP/PAPP has a comparative advantage of being the implementer of the mega waste water infrastructure in Gaza Strip (KY WWTP) and has a deep knowledge and understanding about the project details and different components.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

- **Effectiveness:** UNDP/PAPP worked collaboratively with PENRA to fulfil their aims to achieve 25% of energy demands in the State of Palestine from renewable resources by 2020. In cooperation with PENRA, UNDP implemented various projects in response to the energy crisis in Gaza including the installation of 1.136 MWp PV systems in public schools and heath premises in addition to Rafah treatment and post treatment plants. On the other hand, UNDP/PAPP with the Ministry of Health and World Health Organization conducted in December 2018, a needs assessment on solar energy for 74 health facilities across the Gaza Strip. It comes as part of collective efforts made in the preparation and adaptation of a unified document on solar energy needs for Gaza health facilities.
- The project's cost efficiency and effectiveness will also be enhanced through a variety of approaches, including the following:
    - Following the portfolio management approach, a detailed design for the overall 07 MWp solar energy system for KY WWTP with classified tender documents for the different phases will be provided through the implementation of this 1st phase of the solar energy system, thus, reducing the overall transaction cost required for designing the system in later fragmented intervals/stages.
    - At early stages, UNDP/PAPP intensified efforts with partners and the concerned authorities to acquire land for the solar energy system to be adjacent to KY WWTP site, thus, minimizing the construction costs required for longer electrical cables, extra infrastructure including access roads, etc.
    - Through partnership and leveraging the role of PENRA to participate in the official committee formulated by UNDP for the supervision of the implementation, instead of hiring full time international consultancy firm for that purpose, the transaction cost for monitoring and supervising the project will be reduced.
    - Throughout the design stage, efforts will be done to specify the most advanced technology with highly efficient solar energy panels, inverters, etc., to ensure effective utilization of the investment cost.

Project Management

- The Solar Energy System for Khan Younis Waste Water Treatment Plant (KY WWTP) project will be implemented by UNDP/PAPP in Khan Younis Governorate, Gaza Strip, State of Palestine.
- The Khan Younis Governorate extends over a wide geographic area of around 108 square kilometres in the southern part of the Gaza Strip, and it is inhabited by around 389,000 residents. The Governorate suffered from an absence of a comprehensive sewerage collection system and functional waste water treatment plant. Up to September 2019, the collected raw sewage was discharged into the environment and the Mediterranean Sea; posing serious risks to public health as well as contaminating the ground water aquifer, sea water and marine life.
- With funding from the Government of Japan, the Kuwait Fund for Arab Economic Development through the Islamic Development Bank, and co-financing by UNDP, USD 58 million has been allocated for the construction of KY WWTP, Phase I, with a capacity of 26,600 cubic metres per day. This phase is currently under implementation by UNDP/PAPP. The operation stage of the plant started after completing the commissioning stage in early November 2019.
- The proposed approach for implementing the 07 MWp solar energy project for KY WWTP is expected to provide a constant supply of electricity and reduce the overall running cost by around 50-60%, helping to attain a sustainable operation of the plant.
- With funding from Norway, UNDP/PAPP will implement the first phase of the solar energy system for KY WWTP with a capacity of 1.3 MWp that will contribute to around 35% savings of the electricity costs when the plant is operating at half load capacity.
- UNDP/PAPP delivers through the Direct Execution (DEX) Modality and will be the entity responsible and accountable for managing and implementing the project, including the monitoring and evaluation of project interventions and achieving project outputs. Throughout the implementation processes, UNDP/PAPP plans to work with local partners and local human and capital resources to achieve the project's goals. The capacity of potential implementing partners will be assessed through standard UNDP procedures. In addition, UNDP/PAPP will utilize its full technical and financial capacities through the process to ensure quality implementation of proposed interventions.
- Project evaluation and financial auditing will be conducted, in accordance to UNDP applicable policies and procedures, in annual basis or once during the project lifecycle.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

V. RESULTS FRAMEWORK[2]

Intended Outcome as stated in the UNDAF/Country [or Global/Regional] Programme Results and Resource Framework:
Palestine's infrastructure, and natural and cultural resources are more sustainably used and managed.

Outcome indicators as stated in the Country Programme [or Global/Regional] Results and Resources Framework, including baseline and targets:
Indicator: Percentage of MWp of renewable solar energy needs for KYWWTP integrated within the electricity network to sustainably operate the wastewater treatment plant.
Bassline: 0% solar system in eastern Khan Younis governorate integrated within and feeding the public utility network.
Target: 100% (07/07 MWp) solar system integrated within and feeding the public utility network based on commercial agreement with GEDCO.

Indicator: Amount of wastewater treated daily before reuse for irrigation or discharged into the sea.
Bassline: 0 cubic meters (CM) treated in KYWWTP (+15,000 CM of partially treated waste water discharged daily into the sea).
Target: >15,000 cubic meters are treated daily in KYWWTP.

Applicable Output(s) from the UNDP Strategic Plan: OUTPUT 1.5.1 Solutions adopted to achieve universal access to clean, affordable and sustainable energy

Project title and Atlas Project Number: Solar Energy System for Khan Younis Waste Water Treatment Plant; Project Number: PAL 10: 00117719

EXPECTED OUTPUTS	OUTPUT INDICATORS[3]	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)				DATA COLLECTION METHODS & RISKS
			Value	Year	Year 2020	Year 2021	Year 2022	FINAL	
Output 1 07 MWp Renewable solar energy installed in KYWWTP to sustainably treat the wastewater in Khan Younis governorate.	***1.1*** Capacity of renewable solar energy installed for the sustainable operation of KY WWTP.	UNDP, PWA, PENRA, Khan Younis Municipality, CMWU and GEDCO.	No solar energy source serving KY WWTP.	2019	1.3 MWp solar energy system installed and operated (Phase I).	3.6 MWp solar energy system installed and operated (Phase 2)	2.1 MWp solar energy system installed and operated (Phase 3)	07 MWp solar energy system installed and operated	Monitoring of the system implementation; Evaluation of the progress.

[2] UNDP publishes its project information (indicators, baselines, targets and results) to meet the International Aid Transparency Initiative (IATI) standards. Make sure that indicators are S.M.A.R.T. (Specific, Measurable, Attainable, Relevant and Time-bound), provide accurate baselines and targets underpinned by reliable evidence and data, and avoid acronyms so that external audience clearly understand the results of the project.

[3] It is recommended that projects use output indicators from the Strategic Plan IRRF, as relevant, in addition to project-specific results indicators. Indicators should be disaggregated by sex or for other targeted groups where relevant.

Output 2 Improved awareness of Khan Younis residents especially women regarding the importance of clean water and hygiene, and increased engagement of communities in plans for effective utilization of treated water.	***2.1*** Level of knowledge, attitude and practice among women of Khan Younis governorate on the best practices of hygiene and water-related issues.	Local CBOs UNDP M&E focal point.	Current level of knowledge, attitude and practices.	2019	Improved knowledge, attitude and practices on the best practices of hygiene and water-related issues among girls and women.	Improved knowledge, attitude and practices on the best practices of hygiene and water-related issues among girls and women.		Improved knowledge, attitude and practices on the best practices of hygiene and water-related issues among girls and women.	surveys through questionnaires, interviews with stakeholders and community leaders and workshops.
	2.2 Number of residents who participate in clean water and hygiene awareness campaign in Eastern KY Governorates.	Local CBOs UNDP M&E focal point.	No residents participate in clean water and hygiene awareness campaign in Eastern KY Governorates. Messages formulated to residents on the importance of sustaining the operation of the constructed facilities through their contributions in the water and waste water bills.	*2019*	>300 farmers participated in clean water and hygiene awareness campaign in Eastern KY Governorates.			>300 farmers participated in clean water and hygiene awareness campaign in Eastern KY Governorates.	surveys through questionnaires, interviews with stakeholders and community leaders and workshops.

	2.3 Number of consultations conducted on the effective utilization of the treated water with key stakeholders.	Local CBOs UNDP M&E focal point.	No consultation was conducted on the utilization of treated water of KY WWTP after the designed was finalized in 2010	*2019*	At least one workshop will be conducted with main stakeholder on the effective utilization of the treated water of KY WWTP.	At least one workshop will be conducted with main stakeholder on the effective utilization of the treated water of KY WWTP.		At least two workshops will be conducted with main stakeholder on the effective utilization of the treated water of KY WWTP.	Consultations meetings and workshop with stakeholders.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

VI. MONITORING AND EVALUATION

In accordance with UNDP's programming policies and procedures, the project will be monitored through the following monitoring and evaluation plans:

Monitoring Plan

Monitoring Activity	Purpose	Frequency	Expected Action	Partners (if joint)	Cost (if any)
Track results progress	Progress data against the results indicators in the RRF will be collected and analysed to assess the progress of the project in achieving the agreed outputs.	Semi Annually, or in the frequency required for each indicator	Slower than expected progress will be addressed by project management.		
Monitor and Manage Risk	Identify specific risks that may threaten achievement of intended results. Identify and monitor risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per UNDP's Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage financial risk.	Semi Annually	Risks are identified by project management and actions are taken to manage risk. The risk log is actively maintained to keep track of identified risks and actions taken.		
Learn	Knowledge, good practices and lessons will be captured regularly, as well as actively sourced from other projects and partners and integrated back into the project.	At least annually	Relevant lessons are captured by the project team and used to inform management decisions.		
Annual Project Quality Assurance	The quality of the project will be assessed against UNDP's quality standards to identify project strengths and weaknesses and to inform management decision making to improve the project.	Annually	Areas of strength and weakness will be reviewed by project management and used to inform decisions to improve project performance.		
Review and Make Course Corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	Semi Annually, or at least annually	Performance data, risks, lessons and quality will be discussed by the project board and used to make course corrections.		
Project Report	A progress report will be presented to the Project Board and key stakeholders, consisting of progress data showing the results achieved against pre-defined annual targets at the output level, the annual project quality rating summary, an updated risk long with mitigation measures, and any evaluation or review reports prepared over the period.	Semi Annually, and at the end of the project (final report)			

Project Review (Project Board)	The project's governance mechanism (i.e., project board) will hold regular project reviews to assess the performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the project's final year, the Project Board shall hold an end-of project review to capture lessons learned and discuss opportunities for scaling up and to socialize project results and lessons learned with relevant audiences.	Semi Annually, or at least annually	Any quality concerns or slower than expected progress should be discussed by the project board and management actions agreed to address the issues identified.		

Evaluation Plan[4]

Evaluation Title	Partners (if joint)	Related Strategic Plan Output	UNDAF/CPD Outcome	Planned Completion Date	Key Evaluation Stakeholders	Cost and Source of Funding
Final Evaluation				November 2020 (phase 1) December 2021 (phase 2) December 2022 (phase 3)	PWA, PENRA, GEDCO, CMWU, Khan Younis Municipality, Donor	US$ 6,000 (phase 1) US$ 15,000 (phase 2) US$ 10,000 (phase 3)

[4] Optional, if needed

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

VII. MULTI-YEAR WORK PLAN [56]

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year				RESPONSIBLE PARTY	PLANNED BUDGET 2020 -2022		
		2019	2020	2021	2022		Funding Source	Budget Description	Total Amount
Output 1: 07 MWp Renewable solar energy installed in KYWWTP to sustainably treat the wastewater in Khan Younis governorate. *Gender marker:* GEN 1	1.1 Prepare a comprehensive design package of renewable energy for KY WWTP of 07 MW.	0	120,000.00	0.00	0.00	UNDP	Norway, Other donors		120,000.00
	1.2 Supply and install an on-grid solar energy system including carrying out half year operation after commissioning to exchange knowledge to operator.	0	1,393,467.00	3,800,000.00	2,140,000.00	UNDP	Norway, Other donors		7,333,467.00
	1.3 Conduct training on the operation of the solar energy system.	0	0	0	0	Contractor	Norway, Other donors	Included within activity 1.2	0
	1.4 Construct minimum infrastructure including levelling, fence, transformers & invertors rooms, interlock roads, etc	0	221,710.00	450,000.00	175,000.00	UNDP	Norway, Other donors		846,710.00
	1.5 Obtaining the official documentation by PA for land acquisition and operation.	0	0	0	0	UNDP, PWA, Khan Younis Municipality		No cost required	0
	Sub-Total for Output 1								**8,300,177.00**

[5] Cost definitions and classifications for programme and development effectiveness costs to be charged to the project are defined in the Executive Board decision DP/2010/32

[6] Changes to a project budget affecting the scope (outputs), completion date, or total estimated project costs require a formal budget revision that must be signed by the project board. In other cases, the UNDP programme manager alone may sign the revision provided the other signatories have no objection. This procedure may be applied for example when the purpose of the revision is only to re-phase activities among years.

Output 2: Improved awareness of Khan Younis residents especially women regarding the importance of clean water and hygiene, and increased engagement of communities in plans for effective utilization of treated water. *Gender marker:* GEN 2	2.1 Conduct awareness raising for women in KY governorate on the best practices of hygiene and water -related issues.	0	24,000.00	35,000.00		UNDP	Norway, Other donors		59,000.00
	2.2 Conduct clean water and hygiene awareness raising activities for KY residents, including importance of payment for services.	0	24,000.00			UNDP	Norway, Other donors		24,000.00
	2.3 Conduct consultation session with key stakeholders on effective utilization of treated waste water.	0	4,338.31	15,000.00		UNDP	Norway, Other donors		19,338.31
	Sub-Total for Output 2								**102,338.31**
Monitoring including Evaluation	MONITORING AND EVALUATION	0	210,868.00	283,791.71	270,408.41	UNDP	Norway, Other donors		**765,068.12**
General Management Support (8%)		0	159,870.66	366,703.34	206,832.67	UNDP	Norway, Other donors		**733,406.67**
Coordination Levy (1%)		0	21,582.54	49,504.95	27,922.41	UNDP	Norway, Other donors		**99,009.90**
TOTAL									**10,000,000**

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

- UNDP/PAPP is the Executing Entity of the entire project. The Executing Entity is the entity responsible and accountable for managing and implementing the project, including the monitoring and evaluation of project interventions and achieving project outputs. UNDP/PAPP will carry out the overall and day-to-day management and decision-making for the project and will ensure that the project produces the specified results to the required standard of quality and within the specified constraints of time and cost. UNDP/PAPP roles include the following:
  - Establish and manage the project, while ensuring UNDP's accountability to achieve project outputs.
  - Directly implement and supervise the project activities in close coordination with partners.
  - Recruit consultancy firm to carry out the design and provide the tender documents for the project.
  - Follow global UNDP procurement rules and regulations. (This project will be tendered internationally. All tender documents will be available for international companies in the Global UNDP intranet website, the UNITED NATIONS Global Market (UNGM) website as well as the United Nations Development Business (UNDB) to ensure access to information to all international interested companies.)
  - In full cooperation with PENRA, evaluate the received tenders and award the contract to the most responsive bidder all according to UNDP procurement rules and regulations.
  - Assign a Programme Analyst/ Programme Manager to monitor and follow-up on all managerial issues, including administrative and financial aspects related to the project, and ensure the quality of the project throughout the implementation process.
  - Assign a project team of site and electrical engineers who will be responsible for day-to-day supervision and monitoring of the technical aspects of the project.
  - Prepare the project overall work-plans, ensure proper documentation and provide regular progress reports. The Programme Analyst/ Programme Manager draws up the project plans that describe what the project team will be doing and when they expect to finish.
  - Provide monitoring, quality assurance and quality control of the implemented activities and use of fund in accordance with rules and regulations of UNDP, maintain separate financial records, and submit semi-annual and annual progress and financial reports to the donor.
  - Submit the request for fund disbursement to the donors in line with the progress of the work and disbursements, and assure the funds are utilized as per the Project Document.
  - Provide communication and visibility for the programme.
  - Activate an Issue Log and regularly update it to facilitate tracking and resolution of potential problems or required change.
- The Programme Analyst/ Programme Manager reports to the Project Board on regular basis and is responsible to keep them informed of progress and highlighting any foreseen risks. The Board in its turn approves project plans and offers oversight using management by exception, which means that they intervene in project implementation when exceptions to the approved plan are deemed necessary.

  The Project Board consists of the following roles:

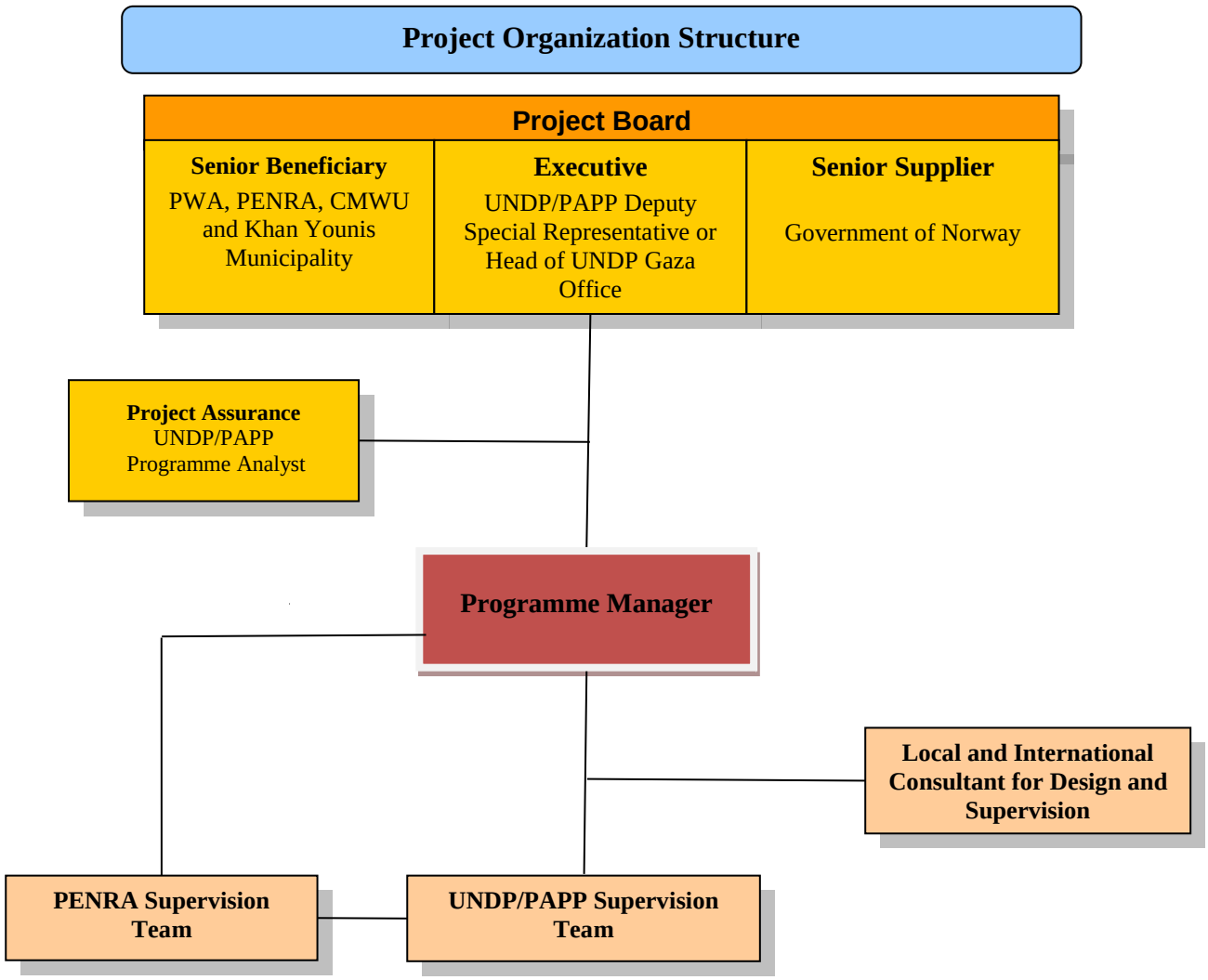
  - Senior Beneficiary: to ensure the realization of project benefits from the perspective of project beneficiaries. This role will be represented by PWA, PENRA, CMWU and Khan Younis Municipality.
  - Senior Supplier: represents that main provider of guidance and contribution regarding the financial feasibility of the project (Norway).
  - Executive: representing the entity accountable for the achievement of results and represented by the UNDP/PAPP Deputy Special Representative.

  The Project Board is composed of the Government of Norway as the senior supplier, UNDP/PAPP as the executive, and the Palestinian Authority represented by PWA, PENRA, CMWU and Khan Younis Municipality as the senior beneficiary. They are organized and can meet semi-annually to jointly coordinate and oversee to ensure that the project delivers the required outputs within budget, on time and to the appropriate quality.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

The Project Board will be supported by the Assurance function. Overall Assurance for this project will be the delegated responsibility of UNDP Programme Analyst. The Project Assurance role supports the Project Board by carrying out regular objective and independent project oversight and monitoring functions. It ensures that appropriate project management milestones are managed and completed. This function will help the board to check that the project remains viable in terms of costs and benefits, that the beneficiaries' requirements are being met, and that the project is delivering suitable solutions on the ground.

The roles and responsibilities of the different partners are outlined in section III/ Partnership.



DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

IX. LEGAL CONTEXT

Option b. Where the country has NOT signed the Standard Basic Assistance Agreement (SBAA)

The project document shall be the instrument envisaged and defined in the Supplemental Provisions to the Project Document, attached hereto and forming an integral part hereof, as "the Project Document".

This project will be implemented by UNDP/PAPP in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

X. RISK MANAGEMENT

Option b. UNDP (DIM)

1. UNDP as the Implementing Partner will comply with the policies, procedures and practices of the United Nations Security Management System (UNSMS.)

2. UNDP as the Implementing Partner will undertake all reasonable efforts to ensure that none of the [project funds][7] [UNDP funds received pursuant to the Project Document][8] are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

3. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

4. UNDP as the Implementing Partner will: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

5. In the implementation of the activities under this Project Document, UNDP as the Implementing Partner will handle any sexual exploitation and abuse ("SEA") and sexual harassment ("SH") allegations in accordance with its regulations, rules, policies and procedures.

6. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

7. UNDP as the Implementing Partner will ensure that the following obligations are binding on each responsible party, subcontractor and sub-recipient:

    a. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of each responsible party, subcontractor and sub-recipient and its personnel and property, and of UNDP's property in such responsible party's, subcontractor's and sub-recipient's custody, rests with such responsible party, subcontractor and sub-recipient. To this end, each responsible party, subcontractor and sub-recipient shall:

[7] To be used where UNDP is the Implementing Partner

[8] To be used where the UN, a UN fund/programme or a specialized agency is the Implementing Partner

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

i. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
ii. assume all risks and liabilities related to such responsible party's, subcontractor's and sub-recipient's security, and the full implementation of the security plan.

b. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the responsible party's, subcontractor's and sub-recipient's obligations under this Project Document.

c. In the performance of the activities under this Project, UNDP as the Implementing Partner shall ensure, with respect to the activities of any of its responsible parties, sub-recipients and other entities engaged under the Project, either as contractors or subcontractors, their personnel and any individuals performing services for them, that those entities have in place adequate and proper procedures, processes and policies to prevent and/or address SEA and SH.

d. Each responsible party, subcontractor and sub-recipient will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, subcontractors and sub-recipients in implementing the project or programme or using the UNDP funds. It will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

e. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to each responsible party, subcontractor and sub-recipient: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. Each responsible party, subcontractor and sub-recipient agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

f. In the event that an investigation is required, UNDP will conduct investigations relating to any aspect of UNDP programmes and projects. Each responsible party, subcontractor and sub-recipient will provide its full cooperation, including making available personnel, relevant documentation, and granting access to its (and its consultants', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with it to find a solution.

g. Each responsible party, subcontractor and sub-recipient will promptly inform UNDP as the Implementing Partner in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

Where it becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, each responsible party, subcontractor and sub-recipient will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). It will provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

h. UNDP will be entitled to a refund from the responsible party, subcontractor or sub-recipient of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of this Project Document. Such amount may be deducted by UNDP from any payment due to the responsible party, subcontractor or sub-recipient under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail any responsible party's, subcontractor's or sub-recipient's obligations under this Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

i. Each contract issued by the responsible party, subcontractor or sub-recipient in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal,

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from it shall cooperate with any and all investigations and post-payment audits.

j. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project or programme, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

k. Each responsible party, subcontractor and sub-recipient shall ensure that all of its obligations set forth under this section entitled “Risk Management” are passed on to its subcontractors and sub-recipients and that all the clauses under this section entitled “Risk Management Standard Clauses” are adequately reflected, *mutatis mutandis*, in all its sub-contracts or sub-agreements entered into further to this Project Document.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

XI. ANNEXES

1. Project Quality Assurance Report

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

2. Social and Environmental Screening Template

Project Information

Project Information	
1. Project Title	Solar Energy System for Khan Younis Waste Water Treatment Plant (KY WWTP)
2. Project Number	Award: PAL 10-00121913, Output: PAL 10-00117719
3. Location (Global/Region/Country)	State of Palestine, Gaza Strip

Part A. Integrating Overarching Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Overarching Principles in order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the Project mainstreams the human-rights based approach
Access to safe, clean water and hygienic sanitation is a basic human right as it was recognized by the United Nations General Assembly on 28 July 2010. Khan Younis Governorate has been sustaining for a long time an absolute absence of a functional wastewater treatment plant. The raw sewage, had been disposed off in the environment and in the Mediterranean Sea without treatment, posing serious risks to public's health as well as contaminating the water aquifer, sea water and marine life, until the KY WWTP was constructed and officially started operation in November 2019. The installation of solar energy system for the operation of KY WWTP project aims at realizing the basic human rights of the 389,000 residents of Khan Younis Governorate for having safe access to safe waste water management. The project includes an awareness and advocacy component which will benefit men and women as right holders at different community settings to raise their awareness of the best practices of hygiene and water-related issues including promotion of their entitlement to access to safe, clean water as a basic human right. Meanwhile, the project supports the PWA as duty-bearers to fulfill their commitments in improving the conditions of wastewater management in the Gaza Strip.
Briefly describe in the space below how the Project is likely to improve gender equality and women's empowerment
The project aims at Improving access to adequate, inclusive and equitable sanitation and hygiene for residents of Khan Younis Governorate with special attention to the needs of women though the optimal operation of the Khan Younis Waste Water Treatment Plant (KYWWTP). One of two main outputs was designed to support gender equality and women's empowerment. Output 2 states: Improved awareness of Khan Younis residents especially women regarding the importance of clean water and hygiene, and increased engagement of communities in plans for effective utilization of treated water. Further, women will also be empowered by minimizing the risks of the water-borne diseases on them and their families, especially children, and the related socio-economic impacts.
Briefly describe in the space below how the Project mainstreams environmental sustainability
The project will utilize the solar energy which is a renewable clean source of energy, thus contributing to preserving the environment by lowering Co2 emissions and safeguarding the aquifer. It also contributes to increasing access to environmentally safe wastewater management system in the Gaza Strip and will minimize pollution in environment, water aquifer, sea water and marine life.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Describe briefly potential social and environmental risks identified in Attachment 1 – Risk Screening Checklist (based on any "Yes" responses). If no risks have been identified in Attachment 1 then note "No Risks Identified" and skip to Question 4 and Select "Low Risk". Questions 5 and 6 not required for Low Risk Projects.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5 below before proceeding to Question 6*			**QUESTION 6: What social and environmental assessment and management measures have been conducted and/or are required to address potential risks (for Risks with Moderate and High Significance)?**
Risk Description	***Impact and Probability (1-5)***	***Significance (Low, Moderate, High)***	***Comments***	***Description of assessment and management measures as reflected in the Project design. If ESIA or SESA is required note that the assessment should consider all potential impacts and risks.***
Inability /incapability of the operator to resume its duties due to the bad economic situation.	I = 4 P = 2	Moderate	The operation of the plant could be interrupted or stopped due to inability of the operator to cover the operation and maintenance cost due to low revenue/collection rates from residents linked with bad economic situation and high unemployment rates in Gaza Strip.	➢ UNDP/PAPP work closely with PWA to find a viable solution ➢ Follow up with PA to sign the commercial agreement between PENRA and GEDCO after the announcement of PWA that PA will cover electricity costs. ➢ Coordinate with Municipality of Khan Younis to secure staff for the plant serving the operator
The design doesn't respect the extent of private properties. The contracting firm doesn't respect the right of way and labor rights.	I =2 P = 1	Low		➢ The design will include instructions to respect human rights in all design aspects. ➢ The identified contracting firm will be required to preserve all human rights during implementation
No participation of women in the awareness sessions. Women engineers lack the opportunity to be trained on solar energy projects.	I = 1 P = 1	Low		➢ Awareness sessions will include women to understand all aspects of KY WWTP project on equal basis as the men. ➢ Training sessions will be conducted for female engineers during the construction stages to enhance their participation in the local market.
The KY WWTP is not operational and the sewage in KY is either percolated into pits or dumped semi treated or untreated into the sea. The construction process results in pollution to air and land.	I = 4 P = 1	Low		➢ Implementation/operation of the project utilizing sustainable energy source will have positive results towards achieving the project environmental goals. ➢ During construction, the contractor will make all measures to minimize the noise and dust pollution in the

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

				area

	QUESTION 4: What is the overall Project risk categorization?		
	Select one (see SESP for guidance)		**Comments**
	Low Risk	X	The project is not likely to involve significant social and environmental risks.
	Moderate Risk	☐	
	High Risk	☐	
	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are relevant?		
	Check all that apply		**Comments**
	Principle 1: Human Rights	X	
	Principle 2: Gender Equality and Women's Empowerment	X	
	1. Biodiversity Conservation and Natural Resource Management	☐	
	2. Climate Change Mitigation and Adaptation	☐	
	3. Community Health, Safety and Working Conditions	X	
	4. Cultural Heritage	☐	
	5. Displacement and Resettlement	☐	
	6. Indigenous Peoples	☐	
	7. Pollution Prevention and Resource Efficiency	X	

Final Sign Off

Signature	***Date***	***Description***
QA Assessor		UNDP staff member responsible for the Project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		
Principles 1: Human Rights		**Answer (Yes/No)**
1.	Could the Project lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
2.	Is there a likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups? [9]	No
3.	Could the Project potentially restrict availability, quality of and access to resources or basic services, in particular to marginalized individuals or groups?	No
4.	Is there a likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them?	No
5.	Is there a risk that duty-bearers do not have the capacity to meet their obligations in the Project?	Yes
6.	Is there a risk that rights-holders do not have the capacity to claim their rights?	No
7.	Have local communities or individuals, given the opportunity, raised human rights concerns regarding the Project during the stakeholder engagement process?	No
8.	Is there a risk that the Project would exacerbate conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Principle 2: Gender Equality and Women's Empowerment		
1.	Is there a likelihood that the proposed Project would have adverse impacts on gender equality and/or the situation of women and girls?	No
2.	Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	No
3.	Have women's groups/leaders raised gender equality concerns regarding the Project during the stakeholder engagement process and has this been included in the overall Project proposal and in the risk assessment?	No
4.	Would the Project potentially limit women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
Principle 3: Environmental Sustainability: Screening questions regarding environmental risks are encompassed by the specific Standard-related questions below		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
1.1	Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	No
1.2	Are any Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3	Does the Project involve changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would	No

[9] Prohibited grounds of discrimination include race, ethnicity, gender, age, language, disability, sexual orientation, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender people and transsexuals.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

	apply, refer to Standard 5)	
1.4	Would Project activities pose risks to endangered species?	No
1.5	Would the Project pose a risk of introducing invasive alien species?	No
1.6	Does the Project involve harvesting of natural forests, plantation development, or reforestation?	No
1.7	Does the Project involve the production and/or harvesting of fish populations or other aquatic species?	No
1.8	Does the Project involve significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.9	Does the Project involve utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)	No
1.10	Would the Project generate potential adverse transboundary or global environmental concerns?	No
1.11	Would the Project result in secondary or consequential development activities which could lead to adverse social and environmental effects, or would it generate cumulative impacts with other known existing or planned activities in the area? *For example, a new road through forested lands will generate direct environmental and social impacts (e.g. felling of trees, earthworks, potential relocation of inhabitants). The new road may also facilitate encroachment on lands by illegal settlers or generate unplanned commercial development along the route, potentially in sensitive areas. These are indirect, secondary, or induced impacts that need to be considered. Also, if similar developments in the same forested area are planned, then cumulative impacts of multiple activities (even if not part of the same Project) need to be considered.*	No
Standard 2: Climate Change Mitigation and Adaptation		
2.1	Will the proposed Project result in significant[10] greenhouse gas emissions or may exacerbate climate change?	No
2.2	Would the potential outcomes of the Project be sensitive or vulnerable to potential impacts of climate change?	No
2.3	Is the proposed Project likely to directly or indirectly increase social and environmental vulnerability to climate change now or in the future (also known as maladaptive practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
Standard 3: Community Health, Safety and Working Conditions		
3.1	Would elements of Project construction, operation, or decommissioning pose potential safety risks to local communities?	Yes
3.2	Would the Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	No
3.3	Does the Project involve large-scale infrastructure development (e.g. dams, roads, buildings)?	No
3.4	Would failure of structural elements of the Project pose risks to communities? (e.g. collapse of buildings or infrastructure)	No
3.5	Would the proposed Project be susceptible to or lead to increased vulnerability to earthquakes, subsidence, landslides, erosion, flooding or extreme climatic conditions?	No
3.6	Would the Project result in potential increased health risks (e.g. from water-borne or other vector-borne diseases or communicable infections such as HIV/AIDS)?	No
3.7	Does the Project pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during Project construction, operation, or decommissioning?	No
3.8	Does the Project involve support for employment or livelihoods that may fail to comply with national and international labor standards (i.e. principles and standards of ILO fundamental conventions)?	No
3.9	Does the Project engage security personnel that may pose a potential risk to health and safety of communities and/or individuals (e.g. due to a lack of adequate training or accountability)?	No

[10] In regards to CO2, 'significant emissions' corresponds generally to more than 25,000 tons per year (from both direct and indirect sources). [The Guidance Note on Climate Change Mitigation and Adaptation provides additional information on GHG emissions.]

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

Standard 4: Cultural Heritage		
4.1	Will the proposed Project result in interventions that would potentially adversely impact sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: Projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.2	Does the Project propose utilizing tangible and/or intangible forms of cultural heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
5.1	Would the Project potentially involve temporary or permanent and full or partial physical displacement?	No
5.2	Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	Yes
5.3	Is there a risk that the Project would lead to forced evictions?[11]	No
5.4	Would the proposed Project possibly affect land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
6.1	Are indigenous peoples present in the Project area (including Project area of influence)?	N/A
6.2	Is it likely that the Project or portions of the Project will be located on lands and territories claimed by indigenous peoples?	N/A
6.3	Would the proposed Project potentially affect the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the Project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to the screening question 6.3 is "yes" the potential risk impacts are considered potentially severe and/or critical and the Project would be categorized as either Moderate or High Risk.*	N/A
6.4	Has there been an absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	N/A
6.5	Does the proposed Project involve the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	N/A
6.6	Is there a potential for forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources?	N/A
6.7	Would the Project adversely affect the development priorities of indigenous peoples as defined by them?	N/A
6.8	Would the Project potentially affect the physical and cultural survival of indigenous peoples?	N/A
6.9	Would the Project potentially affect the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices?	N/A
Standard 7: Pollution Prevention and Resource Efficiency		
7.1	Would the Project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	No
7.2	Would the proposed Project potentially result in the generation of waste (both hazardous and non-hazardous)?	No
7.3	Will the proposed Project potentially involve the manufacture, trade, release, and/or use of hazardous chemicals and/or materials? Does the Project propose use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Stockholm*	No

[11] Forced evictions include acts and/or omissions involving the coerced or involuntary displacement of individuals, groups, or communities from homes and/or lands and common property resources that were occupied or depended upon, thus eliminating the ability of an individual, group, or community to reside or work in a particular dwelling, residence, or location without the provision of, and access to, appropriate forms of legal or other protections.

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

	Conventions on Persistent Organic Pollutants or the Montreal Protocol	
7.4	Will the proposed Project involve the application of pesticides that may have a negative effect on the environment or human health?	No
7.5	Does the Project include activities that require significant consumption of raw materials, energy, and/or water?	No

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

3. Risk Analysis

Project Title: Solar Energy System for Khan Younis Waste Water Treatment Plant				Project Number: PAL 10- 00117719	Date: December 2019
#	**Description**	**Risk Category**	**Impact & Likelihood = Risk Level**	**Risk Treatment / Management Measures**	**Risk Owner**
1	Inability of PWA and Khan Younis Municipality to acquire the land.	Political Strategic Operational	Overall project delayed or cancelled. Scope of work reduced. Partial achievement of results. P = 1, I = 2 L = Low	➢ Intensive coordination between partners and concerned authorities to accelerate acquisition of land. ➢ Consideration for changing the location of the solar energy system to available land inside the plant.	PWA, Khan Younis Municipality, UNDP
2	Continuous supply of power from the PV system to the grid during times of grid outage.	Strategic Operational Financial	Inefficient operation of the solar energy system. The plant operation restricted. Partial achievement of results. P = 1, I = 2 L = Low	➢ GEDCO to ensure continuous availability of electricity (24/7) in the main line of grid connected with the solar energy system. ➢ Study the approach for islanding/micro grid-operational opportunities in the detail design stage.	PWA, PENRA, GEDCO, UNDP
3	Sabotage to the solar PV surface.	Safety and Security Financial Operational	The operation of the plant interrupted. P = 1, I = 1 L = Low	➢ The location of installing PV system was approved by the Israeli authorities. ➢ Contingency and maintenance plan will be activated upon any sabotage incidents.	PWA, Khan Younis Municipality/ CMWU
4	Inability /incapability of the operator to resume its duties due to the bad economic situation.	Strategic Financial Operational Social and Environmental	The operation of the plant interrupted or stopped. Pollution of environment and sea water increased. P = 2, I = 4 L = Medium	➢ UNDP/PAPP work closely with PWA to find a viable solution. ➢ Follow up with PA to sign the commercial agreement between PENRA and GEDCO after the announcement of PWA that PA will cover electricity costs. ➢ Coordinate with Municipality of Khan Younis to secure staff for the plant serving the operator.	PWA, Khan Younis Municipality, UNDP
5	Israeli Military Operation on Gaza.	Political Safety and Security Financial	Overall project delayed or cancelled. Force majeure claims raised by contractors. P = 2, I = 4 L = Medium	➢ Coordination with the Israeli concerned authorities, donors and partners.	UNDP, PWA, PENRA

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

6	Difficulties in access required equipment and construction materials to Gaza Strip.	Political Strategic Financial	Essential components delayed or not implemented. Claims raised by contractors. P = 1, I = 2 L = Low	➢ Coordination with the Israeli concerned authorities, donors and partners. Contingency plan reviewed and developed.	UNDP, PWA, PENRA
7	The design doesn't respect the extent of private properties. The contracting firm doesn't respect the right of way and labour rights.	Social and Environmental Financial	Project progress delayed. Claims raised by residents. P = 1, I = 2 L = Low	➢ The design will include instructions to respect human rights in all design aspects. ➢ The identified contracting firm will be required to preserve all human rights during implementation.	UNDP
8	No participation of women in the awareness sessions. Women engineers lack the opportunity to be trained on solar energy projects.	Social and Environmental	Partial achievement of results. P = 1, I = 1 L = Low	➢ Awareness sessions will include women to understand all aspects of KY WWTP project on equal basis as the men. ➢ Training sessions will be conducted for female engineers during the construction stages to enhance their participation in the local market.	UNDP
9	The KY WWTP is not operational and the sewage in KY is either percolated into pits or dumped semi treated or untreated into the sea. The construction process results in pollution to air and land.	Strategic Operational Financial Social and Environmental	The electrotechnical equipment depreciated and damaged. High maintenance cost required. Pollution of environment and sea water increased. P = 1, I = 4 L = Low	➢ Implementation/operation of the project utilizing sustainable energy source will have positive results towards achieving the project environmental goals. ➢ During construction, the contractor will make all measures to minimize the noise and dust pollution in the area.	UNDP, PWA, CMWU and Khan Younis Municipality
10	Resources are not used in a transparent and accountable manner.	Regulatory Financial	Reputational damages Conflicts generated and progress delayed Financial consequences. P = 1, I = 1 L = Low	➢ Fully follow and implement UNDP rules and regulation. ➢ Publishing of information on the transparency dashboard.	UNDP
11	Delay of manufacturing, shipping and accessing required construction materials and equipment into Gaza Strip due to expanded COVID-19.	Strategic Operational Financial	Construction of project's essential components stopped or delayed. Claims raised by consultant and contractors. P = 3, I = 4 L = Substantial	➢ Coordination with SM, donors and counterparts to be responded. ➢ Contingency plan reviewed and developed.	UNDP, Donor, PWA, PENRA, Contractor

DocuSign Envelope ID: 64E4089F-F38E-4637-9EC7-CB0129AF50CC

4. **Capacity Assessment:** Results of capacity assessments of Implementing Partner (including HACT Micro Assessment)

   N/A

5. **Project Board Terms of Reference and TORs of key management positions**